



Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Acronyms used in this material (1/2)

- ▣ APT: Arbitrage Pricing Theory
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ IPCA: Instrumented Principal Component Analysis
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MOM: MOMentum factor
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component

Acronyms used in this material (2/2)

- ▣ **PCA: Principal Component Analysis**
- ▣ **RF: Risk-Free rate**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **RMW: Robust Minus Weak (profitability factor)**
- ▣ **ROE: Return On Equity**
- ▣ **SMB: Small Minus Big (size factor)**
- ▣ **SML: Security Market Line**
- ▣ **UMD: Up Minus Down (momentum factor)**
- ▣ **US: United States**

Outline

Motivation

From Markowitz to the CAPM

Estimating Beta

Testing the CAPM

Fama–MacBeth Regressions

Multifactor Models

The Factor Zoo and Multiple Testing

Latent Factors: PCA and IPCA

Factor Investing

Conclusions

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What Is a Factor Model?

Definition

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \dots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k ; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock

Key message of this chapter

- Expected returns should be explained by **exposures to priced risks**, not by the name of the asset; testing this claim is the core of empirical asset pricing

Why Factor Models Matter

- ▣ **Cost of capital:** the return investors require from a firm or project
- ▣ **Performance evaluation:** did a fund beat the market, or did it only carry more risk?
- ▣ **Risk management:** N assets need $N(N + 1)/2$ covariances; K factors reduce this drastically
- ▣ **Portfolio construction:** factor exposures can be bought directly (factor ETFs, Section 8)
- ▣ **Asset pricing:** which risks are rewarded, and why? (textbook treatment: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 5–6)

Data Used in This Chapter

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size $\times$ B/M portfolios	US stocks sorted on two characteristics	1963–2026
US sector ETFs	11 sectors of the S&P 500	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 Romanian stocks and their index	2015–2026

- Factor and portfolio returns: [Kenneth French Data Library](#)
- Prices: repository folder data/market, up to 18 September 2026

The Data

- ▣ **ETFs** used in the chapter
 - ▶ Sector ETFs: XLK (technology), XLF (financials), XLE (energy), XLU (utilities), ...; SPY tracks the S&P 500
- ▣ **B/M**: book-to-market ratio, accounting value of equity divided by its market value
- ▣ **BET**: the benchmark index of the BVB
- ▣ Blue chips used: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL (full names on the chart)
- ▣ ETFs and stocks: dividend- and split-adjusted prices; indices: closing values (Chapter 1)

Excess Returns and the Risk-Free Rate

- **Risk-free rate** $R_{f,t}$: return on an asset with no default risk over the period
 - ▶ US: one-month Treasury bill rate (column RF of the Fama–French files); mean 4.35% a year since 1963
- **Excess return**: $R_{i,t}^e = R_{i,t} - R_{f,t}$, the reward for bearing risk
- **Market excess return** (Mkt-RF): value-weighted US market minus $R_{f,t}$
- Assumption for the BVB: daily $R_{f,t}$ is treated as constant
 - ▶ A constant cancels in covariances, so betas are unchanged; only the intercept shifts
- Regressions on Fama–French factors use simple returns (the factors are simple returns); rolling, BVB and PCA analyses use log returns (Chapter 1)

Mean–Variance Portfolio Choice

- ▣ **Markowitz (1952)**: investors care only about the **mean** μ_p and the **variance** σ_p^2 of portfolio returns
- ▣ With weights w , expected excess returns μ and covariance matrix Σ

$$\mu_p = w^\top \mu, \quad \sigma_p^2 = w^\top \Sigma w$$

- ▣ **Efficient frontier**: portfolios with the lowest variance for each target mean

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, \quad w^\top \mathbf{1} = 1$$

- ▣ **Diversification**: covariances, not variances, dominate the risk of a large portfolio

The Tangency Portfolio and the Sharpe Ratio

- ▣ **Sharpe ratio**: excess return per unit of volatility

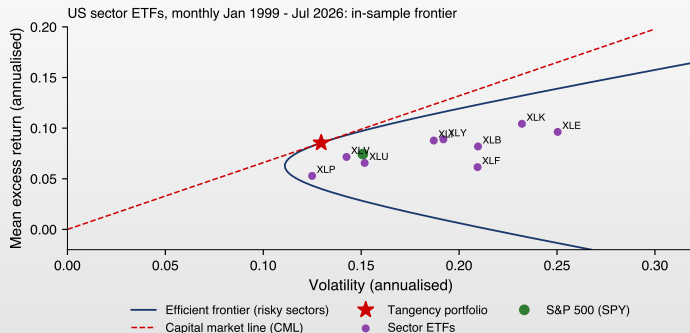
$$SR_p = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p}$$

- ▣ Adding a risk-free asset: the best risky portfolio maximises SR_p ; it is the **tangency portfolio**

$$w_{\text{tan}} = \frac{\Sigma^{-1}\mu}{\mathbf{1}^\top \Sigma^{-1}\mu}$$

- ▣ **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\text{tan}} \sigma_p$
- ▣ Weights can be negative: a **short position** (selling a borrowed asset)

The Efficient Frontier of US Sectors



- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)

Reading the Frontier

- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49
 - ▶ Here the market is **not** on the frontier, as the CAPM would require
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- Caveat: the frontier is **in-sample**; estimated means are very noisy
 - ▶ Out-of-sample optimisation and covariance shrinkage: Chapter 4

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics
1990

Photo: Larry D. Moore (2007); CC BY 4.0;
[Wikimedia Commons](#)

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- Question: if every investor holds a mean–variance efficient portfolio, what are equilibrium prices?
- Answer: everyone holds the same tangency portfolio, so it must be the **market portfolio**
- [Black \(1972\)](#): a version without a risk-free asset (the zero-beta CAPM)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

The CAPM Equation

- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i : sensitivity of asset i to the market
- Main assumptions
 - Mean–variance investors with identical beliefs, one period
 - Unlimited borrowing and lending at R_f ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns

CML versus SML

Capital Market Line (CML)

- ▣ Axes: volatility σ_p vs mean
- ▣ Contains only **efficient** portfolios
- ▣ $\mathbb{E}[R_p] - R_f = SR_m \sigma_p$
- ▣ Single assets lie **below** it

Security Market Line (SML)

- ▣ Axes: beta β_i vs mean
- ▣ Contains **every** asset if the CAPM holds
- ▣ $\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f)$
- ▣ Slope = market premium

- ▣ The SML is the testable statement: plot average returns against betas (Section 4)

Systematic and Idiosyncratic Risk

- Decompose the variance of the market model $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_m^2}_{\text{systematic}} + \underbrace{\sigma_{\varepsilon,i}^2}_{\text{idiosyncratic}}$$

- Systematic risk:** cannot be diversified away; it is rewarded
- Idiosyncratic risk:** vanishes in large portfolios; the CAPM says it earns no premium
- R^2 of the regression $= \beta_i^2 \sigma_m^2 / \sigma_i^2$: the share of systematic risk

Worked Example: CAPM Expected Return

- ▣ Technology ETF XLK, monthly 1999–2026: estimated beta $\hat{\beta} = 1.277$ (Section 3)
- ▣ Historical market premium, 1963–2026: 7.19% a year
- ▣ CAPM expected excess return

$$\mathbb{E}[R_{\text{XLK}}] - R_f = 1.277 \times 7.19\% = 9.18\% \text{ a year}$$

- ▣ With $R_f = 4.35\%$ (1963–2026 average), the required return is $\approx 13.5\%$
- ▣ Both inputs are estimated: the error of the premium dominates the error of beta

Recap: From Markowitz to the CAPM

- Mean–variance investors hold the tangency portfolio combined with the risk-free asset
- In equilibrium this portfolio is the market, and expected excess returns are β_i times the premium
- CML: efficient portfolios in (volatility, mean); SML: every asset in (beta, mean)
- Only systematic risk is rewarded under the CAPM
- In our sector data the market proxy is far from the in-sample frontier

The Market Model Regression

- Time-series regression of asset i on the market, $t = 1, \dots, T$

$$R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$$

- OLS** (Ordinary Least Squares) estimator

$$\hat{\beta}_i = \frac{\sum_t (R_{m,t}^e - \bar{R}_m^e)(R_{i,t}^e - \bar{R}_i^e)}{\sum_t (R_{m,t}^e - \bar{R}_m^e)^2}, \quad \hat{\alpha}_i = \bar{R}_i^e - \hat{\beta}_i \bar{R}_m^e$$

- $\hat{\alpha}_i$: average return not explained by the market (Section 4)
- Classical standard errors assume i.i.d. residuals with constant variance

Robust Standard Errors: Newey–West

- ▣ Returns show volatility clustering (Chapter 1): residuals are **heteroskedastic**
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent covariance, [Newey & West \(1987\)](#)

$$\widehat{\text{Var}}(\hat{b}) = (X^T X)^{-1} \left[\sum_{l=-L}^L w_l \sum_t x_t e_t e_{t-l} x_{t-l}^T \right] (X^T X)^{-1}$$

- ▣ Bartlett weights $w_l = 1 - |l|/(L+1)$; default lag $L = \lfloor 4(T/100)^{2/9} \rfloor$
- ▣ Same point estimates as OLS, only the standard errors change

Worked Example: Beta of the Technology ETF

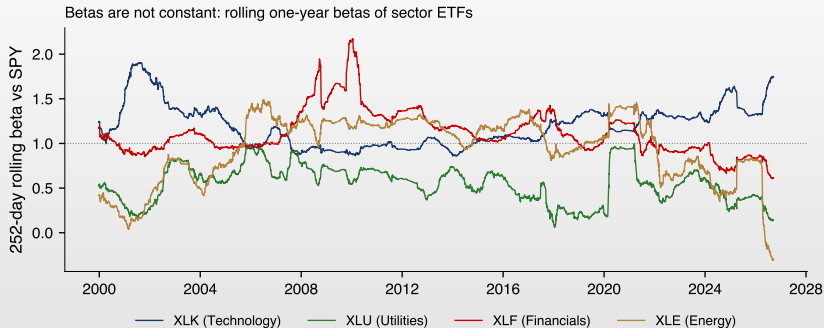
	Estimate	Classical s.e.	Newey–West s.e.
α (annualised)	0.33%	2.27%	2.50%
β	1.277	0.041	0.085

- XLK on the US market factor Mkt-RF, monthly excess returns 1999–2026, $T = 331$, $R^2 = 0.742$; Newey–West with 6 lags
- The robust standard error of beta is **twice** the classical one
- Alpha is statistically zero: $t = 0.33/2.50 = 0.13$

Estimation Choices in Practice

- ▣ **Frequency:** daily data give more observations but suffer from asynchronous trading
- ▣ **Window:** 3–5 years of monthly data or 1–2 years of daily data are common
- ▣ **Market proxy:** S&P 500 (SPY) for US assets, BET for Romanian stocks
- ▣ **Data cleaning:** unadjusted capital events create fake returns
 - ▶ Example: a daily log return of -6.6 for DIGI in May 2017; such days are removed
- ▣ Always report the sample, the frequency and the standard error of beta

Betas Are Not Constant



- Rolling beta: OLS beta on the last 252 daily log returns, recomputed every day

Reading the Rolling Betas

- ▣ Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- ▣ Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end of the sample
- ▣ Energy (XLE): from -0.31 to 1.49; negative at the end, when oil moved against the market
- ▣ Utilities (XLU): from 0.06 to 1.00; the classic defensive sector
- ▣ A single full-sample beta hides these regimes; conditional models: Chapters 5–6

Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is

$$\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$$

- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

- ▣ σ_{β}^2 : cross-sectional variance of betas; noisy estimates are shrunk more

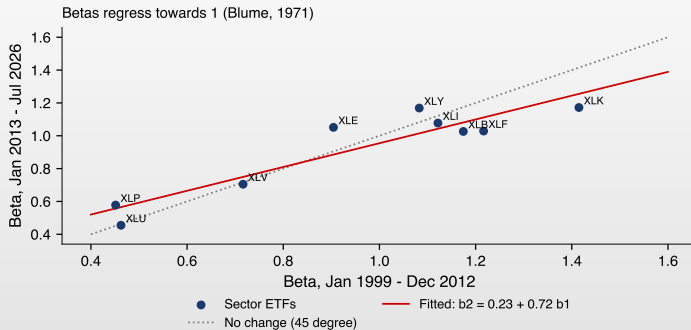
Worked Example: Vasicek Beta of XLK

- ▣ Sample January 1999 – December 2012, monthly, 9 sector ETFs
- ▣ XLK: $\hat{\beta} = 1.415$, $\text{se}(\hat{\beta}) = 0.139$ (Newey–West)
- ▣ Prior: $\bar{\beta} = 0.949$, $\sigma_{\beta}^2 = 0.116$ (mean and variance across the 9 sectors)
- ▣ Weight: $w = 0.116 / (0.116 + 0.139^2) = 0.857$

$$\beta^V = 0.857 \times 1.415 + 0.143 \times 0.949 = 1.348$$

- ▣ The realised beta in 2013–2026 was 1.172: shrinkage moved the estimate in the right direction

Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)

Reading the Beta Regression

- ▣ Fitted line: $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67 \hat{\beta}$
- ▣ **RMSE** (Root Mean Squared Error) of the forecast of later betas

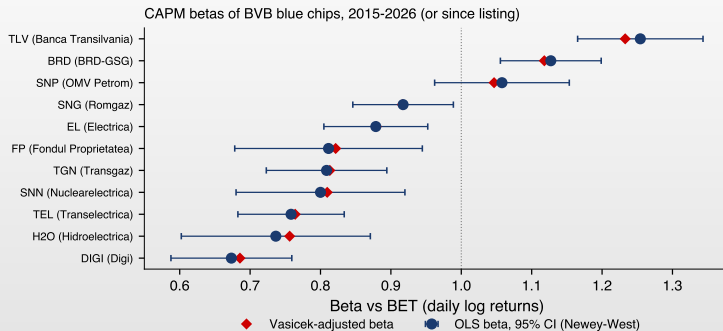
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$$

- ▣ Raw beta: 0.135; Blume rule: 0.109; Vasicek: 0.115
- ▣ Shrinkage cuts the forecast error by about 20%
- ▣ Only 9 sectors: the lesson is qualitative

Romania: Betas of the BVB Blue Chips

- ▣ Regression of daily log returns of each stock on daily log returns of BET
- ▣ Sample from 6 January 2015 or from the listing date (DIGI 2017, H2O 2023)
- ▣ Risk-free rate: assumed constant at daily frequency, so it does not affect beta
- ▣ Days with $|r_t| > 0.5$ (capital events not adjusted in prices) are removed: one day for DIGI, two for FP
- ▣ Newey–West 95% confidence intervals and Vasicek-adjusted betas (prior: the 11 stocks)

CAPM Betas on the Bucharest Stock Exchange



□ Blue dots: OLS betas with 95% intervals; red diamonds: Vasicek-adjusted betas

Reading the BVB Betas

- ▣ Banks are the most market-sensitive: Banca Transilvania 1.25, BRD 1.13
- ▣ OMV Petrom 1.06; Romgaz 0.92; utilities and telecom below: Transelectrica 0.76, Digi 0.67
- ▣ R^2 : 0.59 for TLV and 0.52 for BRD, but only 0.21 for DIGI
- ▣ Large index weights (TLV, SNP, BRD) mechanically raise their R^2 against BET
- ▣ Vasicek changes little here: the daily estimates are precise (se $\approx$ 0.04–0.07)

Recap: Estimating Beta

- Beta is an OLS slope; report it with Newey–West standard errors
- Betas move over time: sector betas ranged from about -0.3 to 2.2 in rolling windows
- Extreme betas regress towards one: Blume and Vasicek shrinkage improve forecasts
- Romanian banks have the highest betas against BET; utilities and telecom the lowest
- Clean the data first: a single unadjusted corporate event destroys a regression

Jensen's Alpha

- ▣ **Jensen's alpha** (Jensen (1968)): intercept of the market model

$$\alpha_i = \mathbb{E}[R_i^e] - \beta_i \mathbb{E}[R_m^e]$$

- ▣ The CAPM implies $\alpha_i = 0$ for every asset
- ▣ $\alpha_i > 0$: the asset earned more than its market risk justifies
- ▣ Test one asset: $t = \hat{\alpha}_i / \text{se}(\hat{\alpha}_i)$
- ▣ Testing many assets one by one inflates false rejections (Section 7): we need a joint test

The GRS Test

- ▣ **GRS:** Gibbons, Ross & Shanken (1989), joint test of $H_0 : \alpha_1 = \dots = \alpha_N = 0$

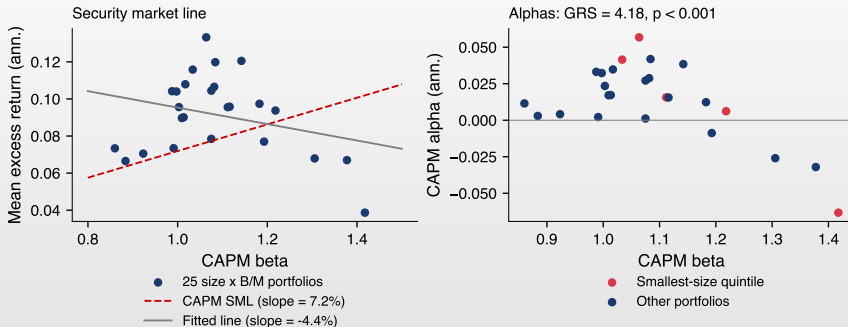
$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣ $\hat{\Sigma}$: residual covariance matrix; $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market
- ▣ Exact under normally distributed residuals; F : Fisher distribution
- ▣ Interpretation: large W means the market is far from the mean–variance frontier of the test assets

The 25 Size and Book-to-Market Portfolios

- ▣ US stocks sorted each June into 5 **size** quintiles (market value) and 5 **B/M** quintiles
- ▣ $5 \times 5 = 25$ value-weighted portfolios: from small-growth (small, low B/M) to big-value
- ▣ Standard test assets since [Fama & French \(1993\)](#): they spread average returns widely
- ▣ Monthly, July 1963 – July 2026, $T = 757$
- ▣ Portfolios instead of single stocks: betas are estimated much more precisely

The Security Market Line in the Data



□ Left: average excess returns vs CAPM betas; right: alphas, smallest-size quintile in red

Reading the Security Market Line

- ▣ The CAPM predicts a line with slope equal to the premium, 7.2% a year
- ▣ The fitted cross-sectional line has slope -4.4% : higher beta, **lower** average return
- ▣ Betas span only 0.86–1.42, while average excess returns span 3.9%–13.3%
- ▣ Small-value alpha $+5.7\%$, small-growth alpha -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- ▣ $GRS = 4.18$, p -value 8×10^{-11} : the CAPM is rejected

A Flat SML: Black's Insight



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage
- High-beta assets become expensive, so their future returns are lower
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above R_f
- The same Black co-authored the Black–Scholes option formula (Chapter 12)

Betting Against Beta

- Frazzini & Pedersen (2014) turn the flat SML into a trading strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L : low-beta portfolio, leveraged to beta 1; H : high-beta portfolio, de-leveraged to beta 1
- The portfolio has beta zero by construction, yet earns a positive average return in their data
- Explanation: leverage constraints of investors, as in Black (1972)

Roll's Critique

- ▣ Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth
- ▣ Stock indices omit bonds, real estate, private firms and human capital
- ▣ Mathematical fact: any mean–variance efficient portfolio prices all assets exactly by beta
- ▣ Hence a CAPM test is a joint test of the model and of the efficiency of the proxy
- ▣ Practical answer: evaluate models by how well they price assets, not as literal truths

Low Power: GRS on Sector ETFs

- ▣ Same test on the 9 sector ETFs, monthly 1999–2026 ($T = 331$)
- ▣ $GRS = 0.53$, $p = 0.85$: the CAPM is **not** rejected
- ▣ Alphas: utilities +3.4%, energy +2.4%, financials −2.3% a year, all statistically weak
- ▣ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- ▣ Lesson: failing to reject is weak evidence; choose test assets with dispersion in expected returns

Recap: Testing the CAPM

- ▣ The CAPM says all alphas are zero; the GRS test checks this jointly
- ▣ On the 25 size $\times$ B/M portfolios, 1963–2026, the CAPM is strongly rejected
- ▣ The empirical SML is flat or negative; leverage constraints (Black; BAB) offer one explanation
- ▣ Roll: every test also tests the market proxy
- ▣ Test assets matter: sector ETFs cannot reject the CAPM

Cross-Sectional Tests: Fama and MacBeth



Eugene Fama receiving the Morgan Stanley–AFA
Award, 2008

Photo: [TonyTheTiger \(2008\)](#); CC BY-SA 3.0;
[Wikimedia Commons](#)

- **Fama & MacBeth (1973)**: estimate the **price of risk** λ of each factor
- **Risk premium** λ_k : extra expected return per unit of exposure $\beta_{i,k}$
- Model: $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$
- Allows factors that are **not** traded portfolios, and firm characteristics as regressors
- Fama: Nobel Prize in Economics 2013

The Two Passes

- ▣ **Pass 1** (time series): for each asset i , estimate the loadings

$$R_{i,t}^e = a_i + \beta_i^\top \mathbf{f}_t + \varepsilon_{i,t} \quad \Rightarrow \quad \hat{\beta}_i$$

- ▣ **Pass 2** (cross section): for each month t , regress returns on the estimated betas

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \quad i = 1, \dots, N$$

- ▣ Estimate: $\hat{\boldsymbol{\lambda}} = \frac{1}{T} \sum_t \hat{\boldsymbol{\lambda}}_t$, the average of T monthly slopes
- ▣ λ_0 : pricing error common to all assets; zero if the model is right

Fama–MacBeth Standard Errors

- Treat the monthly estimates $\hat{\lambda}_{k,t}$ as a sample of size T

$$se_{FM}(\hat{\lambda}_k) = \frac{s(\hat{\lambda}_{k,t})}{\sqrt{T}}, \quad s^2 = \frac{1}{T-1} \sum_t (\hat{\lambda}_{k,t} - \hat{\lambda}_k)^2$$

- Advantage: automatically robust to cross-sectional correlation of residuals
- If $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series instead
- Problem: the betas of Pass 2 are **estimated**, not known

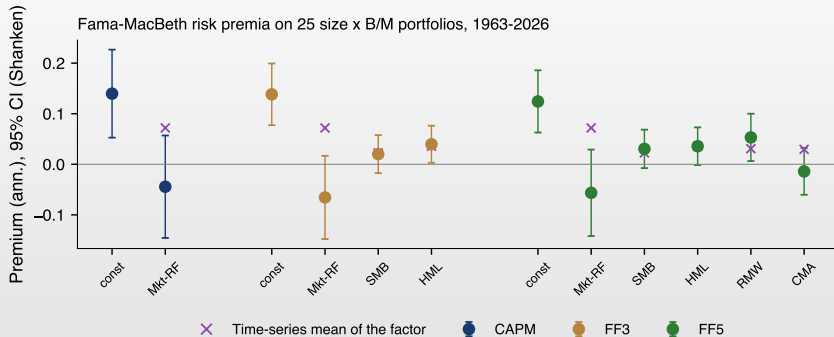
Errors in Variables: the Shanken Correction

- ▣ **Errors in variables:** using $\hat{\beta}$ instead of β makes se_{FM} too small
- ▣ **Shanken (1992)** correction for the factor premia

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \text{Var}_{FM}(\hat{\lambda}) + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

- ▣ Σ_f : covariance matrix of the factors; c : squared Sharpe ratio implied by the premia
- ▣ The correction matters most for monthly data and large premia
- ▣ Equivalent results follow from **GMM** (Generalized Method of Moments, [Hansen \(1982\)](#)); see Quantinar

Estimated Risk Premia



25 size $\times$ B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means

Reading the Risk Premia

- ▣ CAPM: intercept 14.0% (se 4.4%), market premium -4.4% (se 5.2%)
- ▣ FF3: SMB 2.0% (1.9%), HML 4.0% (1.9%); the market premium stays negative
- ▣ FF5: RMW 5.3% (2.4%); CMA -1.4% (2.4%) is not priced in these portfolios
- ▣ For a traded factor, λ should equal the time-series factor mean (the crosses)
 - ▶ HML: 4.0% vs mean 3.6%, close; market: -6.6% vs mean 7.2%, far
- ▣ A large intercept means the test assets are not priced by the model

Worked Example: How Much Does Shanken Matter?

FF3, HML	$\hat{\lambda}$	s.e.	t	Significant at 5%?
Fama–MacBeth	3.96%	1.33%	2.98	yes
Newey–West (6 lags)	3.96%	1.62%	2.44	yes
Shanken	3.96%	1.87%	2.12	yes, barely

- Same point estimate, three standard errors: t falls by 30% after the correction
- With the $t > 3$ hurdle of Section 7, HML would not pass with any of them

Recap: Fama–MacBeth

- ▣ Pass 1 estimates betas; Pass 2 averages monthly cross-sectional slopes
- ▣ Fama–MacBeth errors handle cross-sectional correlation; Newey–West handles autocorrelation
- ▣ Shanken corrects for estimated betas and can change conclusions
- ▣ On the 25 portfolios the market premium is negative, value and profitability are priced
- ▣ Premia of traded factors should match their time-series means

Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a K -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣ The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread, $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French 2×3 sort: size (small/big) × characteristic (30% / 40% / 30%)
 - ▶ **SMB** (Small Minus Big): small stocks minus big stocks
 - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- Portfolios rebalanced each June using accounting data of the previous year

The Fama–French Models

- Fama & French (1992): size and B/M explain the cross section; beta does not
- **FF3** (three-factor model), Fama & French (1993)

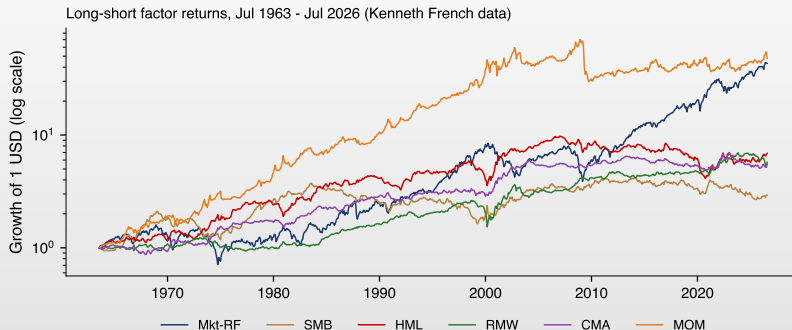
$$R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$$

- **FF5** (five-factor model), Fama & French (2015), adds
 - ▶ **RMW** (Robust Minus Weak): high minus low operating profitability
 - ▶ **CMA** (Conservative Minus Aggressive): low minus high asset growth (investment)
- Test of a model: time-series alphas close to zero, GRS not rejected

Momentum, Carhart and the q-Factor Model

- ▣ **Momentum**: past winners keep winning over 3–12 months, [Jegadeesh & Titman \(1993\)](#)
- ▣ **MOM** (UMD, Up Minus Down): stocks with high minus low return from month $t - 12$ to $t - 2$
- ▣ [Carhart \(1997\)](#) four-factor model: FF3 + MOM, standard for fund evaluation
- ▣ **q-factor model**, [Hou, Xue & Zhang \(2015\)](#): market, size, investment and profitability (ROE)
 - ▶ ROE: Return On Equity; motivated by firms' investment decisions
- ▣ FF5 and the q-model are close relatives; FF5 omits momentum

Six Factors over Sixty-Three Years

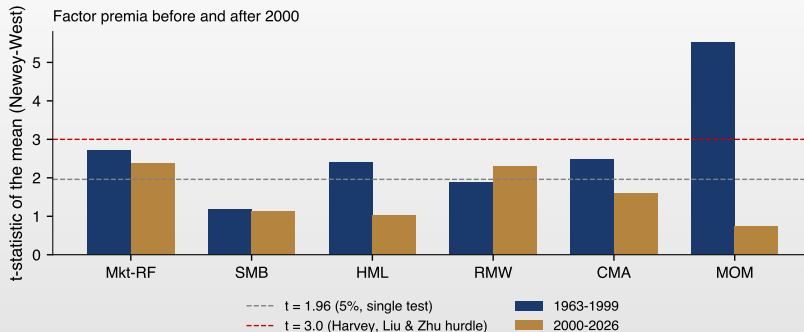


- Growth of 1 USD invested in each long-short factor (no interest on collateral), monthly, log scale

Reading the Factor Growth

- ▣ July 1963 – July 2026: 1 USD grew to 43.2 in Mkt-RF and to 48.0 in MOM
- ▣ Value (HML) 6.9, profitability (RMW) 5.7, investment (CMA) 5.5, size (SMB) 2.9
- ▣ Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- ▣ Value was flat or losing from about 2007 to 2020
- ▣ Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

Do the Premia Survive?



□ t-statistic of the mean monthly factor return (Newey-West), 1963–1999 vs 2000–2026

Reading the Split-Sample t -Statistics

- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31
- Market premium robust in both halves ($t = 2.70$ and 2.39)
- McLean & Pontiff (2016): anomaly returns are 58% lower after academic publication, consistent with this decay

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in
Economics 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0;
[Wikimedia Commons](#)

- **SDF** (Stochastic Discount Factor) m_t : prices all excess returns, $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#)
- Survey of the discount-rate view: [Cochrane \(2011\)](#)
- Hansen shared the 2013 Nobel Prize with Fama and Shiller

Recap: Multifactor Models

- ▣ The APT justifies several priced factors but does not name them
- ▣ Factors are long-short portfolios sorted on characteristics: SMB, HML, RMW, CMA, MOM
- ▣ FF3, Carhart, FF5 and the q-model are the standard benchmarks
- ▣ Momentum and value were much weaker after 2000; profitability stayed strong
- ▣ Factor models are SDFs and can be estimated by GMM

The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs $|t| > 3.0$

Family-Wise Error Rate: Bonferroni and Holm

- ▣ M hypotheses with p -values $p_1, \dots, p_M$; target level α
- ▣ **FWER** (Family-Wise Error Rate): probability of at least one false rejection
- ▣ **Bonferroni**: reject H_j if $p_j \leq \alpha/M$; controls FWER for any dependence
- ▣ **Holm** (Holm (1979)): sort $p_{(1)} \leq \dots \leq p_{(M)}$; reject while

$$p_{(j)} \leq \frac{\alpha}{M - j + 1}$$

- ▣ Holm rejects at least as much as Bonferroni with the same FWER guarantee

False Discovery Rate: Benjamini–Hochberg

- ▣ **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- ▣ **BH** procedure (Benjamini & Hochberg (1995)): find the largest j with

$$p_{(j)} \leq \frac{j}{M} \alpha$$

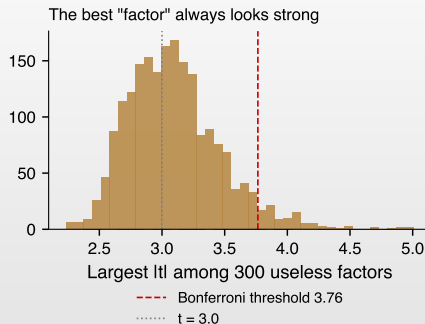
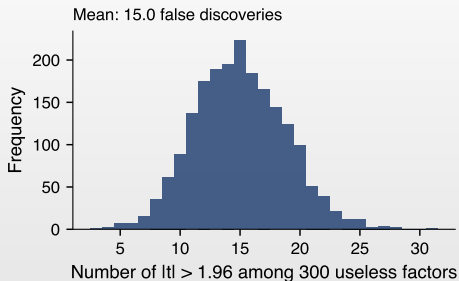
and reject $H_{(1)}, \dots, H_{(j)}$

- ▣ Controls FDR at α for independent (or positively dependent) tests
- ▣ Less strict than FWER control: suited to searching many candidate factors

Worked Example: Ten Candidate Factors

- ▣ Sorted p -values: 0.001, 0.004, 0.009, 0.012, 0.021, 0.035, 0.048, 0.060, 0.20, 0.55; $\alpha = 5\%$
- ▣ Naive ($p \leq 0.05$): **7** discoveries
- ▣ Bonferroni ($p \leq 0.005$): **2**
- ▣ Holm: $0.001 \leq 0.005$, $0.004 \leq 0.0056$, $0.009 > 0.00625$ stop: **2**
- ▣ BH thresholds 0.005, 0.010, 0.015, 0.020, 0.025, $\dots$: $p_{(5)} = 0.021 \leq 0.025$ is the last success: **5**

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t -statistics from the standard Normal distribution (no true premium)

Reading the Simulation

- ▣ On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- ▣ On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications
- ▣ The median largest $|t|$ is 3.06: the best “factor” always looks strong
- ▣ Bonferroni threshold for 300 tests: $|t| > 3.76$
- ▣ Hurdles rise with the number of tests actually tried, including unpublished ones

Taming the Zoo and the Replication Debate

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add information given the existing zoo?
 - ▶ **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator): penalised regressions choose the controls among hundreds of factors
 - ▶ Most new factors are redundant; only a few add pricing information
- [Jensen, Kelly & Pedersen \(2023\)](#): a Bayesian view of 153 factors in 93 countries
 - ▶ Most factors replicate; they cluster into 13 themes; out-of-sample and international evidence is consistent
- The two views agree: the zoo is highly redundant, and a few themes carry the information

Recap: The Factor Zoo

- ▣ Hundreds of published factors imply many false discoveries at $|t| > 1.96$
- ▣ Bonferroni and Holm control the chance of any false discovery; BH controls their share
- ▣ With 300 useless factors, the best one has $|t| \approx 3$: raise the hurdle
- ▣ Redundancy is the main problem: few themes carry most of the information

Principal Component Analysis

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis): eigen-decomposition of the correlation matrix C of standardised returns

$$C = \Gamma \Lambda \Gamma^\top, \quad y_t = \Gamma^\top z_t$$

- ▣ $\Lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_N)$: eigenvalues; Γ : eigenvectors (**loadings**)
- ▣ Share of total variance explained by component k : $\lambda_k / \sum_j \lambda_j = \lambda_k / N$
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components

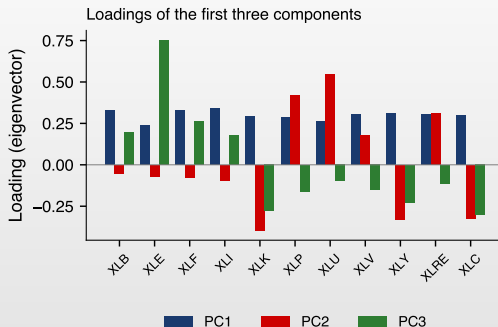
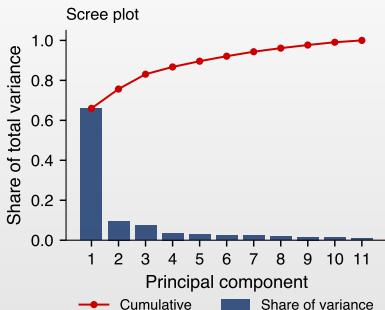
Worked Example: PCA of Two Assets

- Two standardised returns with correlation $\rho = 0.6$

$$C = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 1 \end{pmatrix}, \quad \lambda_1 = 1 + \rho = 1.6, \quad \lambda_2 = 1 - \rho = 0.4$$

- Eigenvectors: $(1, 1)/\sqrt{2}$ (the “market”) and $(1, -1)/\sqrt{2}$ (the spread)
- The first component explains $1.6/2 = 80\%$ of the variance
- Higher correlation, stronger first factor: in a crisis correlations rise and PC1 dominates

PCA of the US Sector ETFs



11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)

Reading the Principal Components

- ▣ PC1 explains 66.0% of the variance; all loadings are positive (0.24–0.34)
 - ▶ Correlation of PC1 with SPY: 0.954; the first latent factor is the market
- ▣ PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology –0.40, consumer discretionary –0.33
 - ▶ A **defensive-versus-growth** factor, with no name given in advance
- ▣ PC3 (7.4%) is dominated by energy; three components explain 83.1%
- ▣ Only 2 eigenvalues exceed 1 (Kaiser rule for the number of factors)

Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Signs and rotations are arbitrary: interpretation requires care
- ▣ PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#); multivariate background: [Härdle & Simar \(2019\)](#)

Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of L observed characteristics $\mathbf{z}_{i,t}$ (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- ▣ $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?

What IPCA Teaches

- Title of the paper: “Characteristics are covariances”
- In their US stock data, a few IPCA factors price the cross section better than observable-factor models with as many factors
- The alpha term Γ_α becomes insignificant: characteristics predict returns because they measure risk exposure
- Links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Related use of dimension reduction on return features of cryptos and traditional assets: [Pele et al. \(2023\)](#)

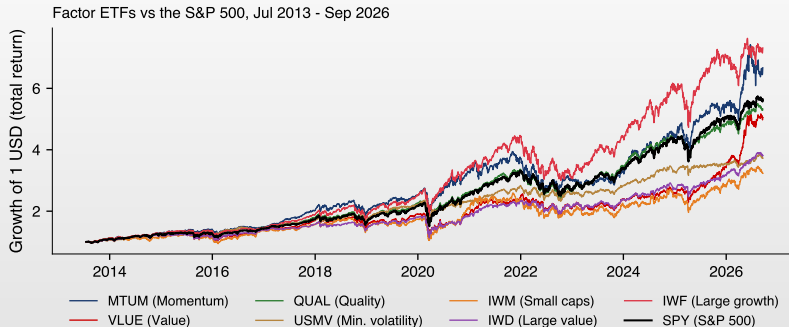
Recap: Latent Factors

- PCA finds factors as eigenvectors of the correlation matrix
- In sector ETFs, PC1 is the market (66%), PC2 a defensive-versus-growth factor (10%)
- PCA explains variance, with fixed loadings; it is not an asset pricing model by itself
- IPCA lets loadings depend on characteristics and tests whether characteristics are risk

Factor ETFs

- ▣ **Smart beta:** rule-based portfolios that tilt towards factors, sold as ETFs
- ▣ MTUM (momentum), VLU (value), QUAL (quality: high profitability, low leverage)
- ▣ USMV (minimum volatility: the low-risk anomaly), IWM (small caps, Russell 2000)
- ▣ IWD / IWF: Russell 1000 value / growth (large caps)
- ▣ Long-only: they hold only the long leg, so their market beta is close to 1

Factor ETFs versus the Market

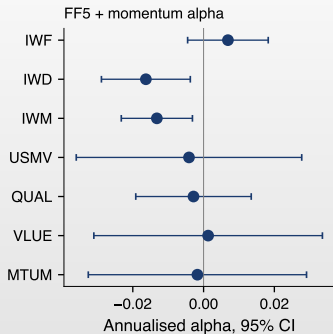


□ Growth of 1 USD with dividends re-invested, from 18 July 2013 (first common day)

Reading the Factor ETF Returns

- ▣ **CAGR** (Compound Annual Growth Rate): $(P_T/P_0)^{1/\text{years}} - 1$
- ▣ Large growth (IWF) 16.3% and momentum (MTUM) 15.5% beat SPY (14.0%)
- ▣ Quality 13.5%, value 13.1%, large value 10.7%, minimum volatility 10.5%, small caps 9.3%
- ▣ This period favoured large growth stocks: value and small caps lagged
- ▣ Thirteen years of one realisation: not evidence against the premia on its own

Alphas of Factor ETFs



Factor loadings

	Mkt-RF	SMB	HML	RMW	CMA	MOM
MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00

□ Daily regressions on FF5 + MOM with Newey–West errors, each ETF on its own trading calendar (2000–2026)

 MFM_ch3_etf_alphas

Reading the ETF Alphas

- ▣ The loadings match the labels: MTUM on MOM 0.36, VLUU on HML 0.34, IWM on SMB 0.85
- ▣ Growth (IWF) loads -0.28 on HML; USMV has market beta 0.74 and positive RMW, CMA
- ▣ Factor-model alphas are close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%
- ▣ Russell ETFs: IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.55$): costs and index construction
- ▣ R^2 between 0.81 and 0.97: the ETFs are mostly repackaged factor exposure

Factor Timing and Crowding

- ▣ **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
 - ▶ Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- ▣ **Crowding:** when many investors hold the same factor positions
 - ▶ Premia can shrink ([McLean & Pontiff \(2016\)](#)) and forced selling can cause crashes, as in momentum in 2009
- ▣ **Costs:** long-short factors ignore trading costs and short-selling fees; ETFs pay them

Recap: Factor Investing

- ▣ Factor ETFs deliver the promised exposures, with market betas near 1
- ▣ After controlling for the factors, their alphas are zero or slightly negative
- ▣ 2013–2026 rewarded growth and momentum, punished value and size
- ▣ Timing and crowding are open problems; evaluate them out of sample

Conclusions

- ▣ The CAPM is elegant but fails on the 25 size $\times$ B/M portfolios: the SML is flat (slope -4.4%), $GRS = 4.18$
- ▣ Betas must be estimated with robust errors, change over time and should be shrunk
- ▣ Fama–MacBeth with Shanken errors: value and profitability are priced, the market premium is not in the cross section
- ▣ Hundreds of factors require multiple-testing hurdles ($|t| > 3$)
- ▣ Latent factors: PCA recovers the market and a defensive-versus-growth factor; IPCA links them to characteristics
- ▣ Factor ETFs are cheap factor exposure, not alpha

Links to Other Chapters

- ▣ **Chapter 2** (market efficiency): an alpha is either an anomaly (inefficiency) or a missing risk factor; the **joint hypothesis** problem
- ▣ **Chapter 4** (portfolio optimisation): factor covariance models and shrinkage for Markowitz
- ▣ **Chapters 5–6**: time-varying volatility and correlation, conditional betas
- ▣ **Chapter 8**: backtesting factor strategies without overfitting
- ▣ **Chapter 13**: machine learning for the cross section of returns

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- ▣ Factor data: [Kenneth French Data Library](#)

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